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Education

Ph.D. in Aerospace Engineering , Georgia Institute of Technology Advisor: Prof. Kyriakos G. Vamvoudakis Thesis: Game-Theoretic Learning and Control for Resilience of Complex Adaptive Systems	2019–2024
M.S. in Mathematics , Georgia Institute of Technology	2019–2023
M.S. in Aerospace Engineering , Georgia Institute of Technology	2019–2022
Diploma (B.S. and M.S.) in Electrical & Computer Engineering , Aristotle University of Thessaloniki Advisor: Prof. George A. Rovithakis Thesis: Improving Robustness of Adaptive Dynamic Programming Controllers using Prescribed Performance Control	2013–2018

Professional & Research Experience

Postdoctoral Fellow , University of Texas at Austin Autonomous Systems Group, PI: Prof. Ufuk Topcu	08/2024–present
Graduate Research Assistant , Georgia Institute of Technology Intelligent Cyber-Physical Systems Lab, PI: Prof. Kyriakos G. Vamvoudakis	08/2019–08/2024

Teaching Experience

Instructor , AE8750/8751: Robotics Research Fundamentals Georgia Institute of Technology	Spring 2023
Guest Lecturer , AE8803: Optimization-Based Learning Control and Games Georgia Institute of Technology	Fall 2022, 2023

Awards & Honors

Reviewer of the Year Award , ASME Journal of Dynamic Systems, Measurement, and Control	2024
Cyber-Physical Systems Rising Star (20% acceptance rate) , National Science Foundation	2024
Georgia Tech Aerospace Engineering Fellowship	2023–2024
Alexander S. Onassis Foundation Scholarship	2020–2024
Outstanding Reviewer Award (top 1%) , IEEE Control Systems Letters	2023
Excellent Reviewer Award , Journal of Guidance, Control, and Dynamics	2022
Gerondelis Foundation Graduate Study Scholarship	2021

Publications

Books

1. A. Kanellopoulos, L. Zhai, **F. Fotiadis**, K. G. Vamvoudakis, “Control and Game Theoretic Methods for Cyber-Physical Security,” *Elsevier*, 2024.

Book Chapters

2. **F. Fotiadis**, L. Zhai, K. G. Vamvoudakis, J. Hugues, “On the Impact of Sensor-Actuator Clock Offsets on Off-Policy Reinforcement Learning,” in *Smarter Cyber-Physical Systems: Enabling Methodologies and Applications*, eds. Y. Wan, K. G. Vamvoudakis, Y. Chen, F. L. Lewis, CRC Press, 2024.

Journal Publications

3. X. Liu, J. Li, **F. Fotiadis**, M. O. Karabag, J. Milzman, D. Fridovich-Keil, U. Topcu, “Approximate Feedback Nash Equilibria with Sparse Inter-Agent Dependencies,” conditionally accepted in *IEEE Transactions on Automatic Control*, 2026.
4. J. Im, **F. Fotiadis**, D. Delahaye, U. Topcu, D. Fridovich-Keil, “Noncooperative Coordination via a Trading-based Auction,” conditionally accepted in *IEEE Transactions on Automatic Control*, 2026.
5. **F. Fotiadis**, A. Kanellopoulos, K. G. Vamvoudakis, U. Topcu, “Deception Against Data-Driven Linear-Quadratic Control,” conditionally accepted in *IEEE Transactions on Automatic Control*, 2026.
6. G. Cooper, N. Tsao, **F. Fotiadis**, U. Topcu, R. Marculescu, “BHAR: A System-Level Optimization Framework for Multisensor Batteryless Human Activity Recognition,” to appear in *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2026.

7. T. Qiu, **F. Fotiadis**, X. Liu, C. Ellis, J. Milzman, W. Suttle, U. Topcu, D. Fridovich-Keil, “Linear-Quadratic Gaussian Games with Distributed Sparse Estimation,” *IEEE Control Systems Letters*, vol. 10, pp. 1237-1242, 2026.
8. **F. Fotiadis**, K. G. Vamvoudakis, “Input-Output Data-Driven Sensor Selection for Cyber-Physical Systems,” *Automatica*, vol. 186, paper 112829, 2026.
9. **F. Fotiadis**, K. G. Vamvoudakis, “A Physics-Informed Learning Framework to Solve the Infinite-Horizon Optimal Control Problem,” *International Journal of Robust and Nonlinear Control*, vol. 35, no. 16, pp. 6932–6944, 2025.
10. S. Niu, A. Bouland, H. Wang, **F. Fotiadis**, A. Kurdila, A. L’Afflitto, S. T. Paruchuri, K. G. Vamvoudakis, “Convergence Rates of Online Critic Value Function Approximation in Native Spaces,” *IEEE Control Systems Letters*, vol. 8, pp. 2145-2150, 2024.
11. **F. Fotiadis**, A. Kanellopoulos, K. G. Vamvoudakis, J. Hugues, “On the Effect of Clock Offsets and Quantization on Learning-based Adversarial Games,” *Automatica*, vol. 167, paper 111762, 2024.
12. **F. Fotiadis**, G. A. Rovithakis, “Input-Constrained Prescribed Performance Control for High-order MIMO Uncertain Nonlinear Systems via Reference Modification,” *IEEE Transactions on Automatic Control*, vol. 69, no. 5, pp. 3301-3308, 2024.
13. **F. Fotiadis**, K. G. Vamvoudakis, Z.-P. Jiang, “Data-Driven Actuator Allocation for Actuator Redundant Systems,” *IEEE Transactions on Automatic Control*, vol. 69, no. 4, pp. 2249-2264, 2024.
14. **F. Fotiadis**, K. G. Vamvoudakis, “Learning-based Actuator Selection for Increased Attack Resilience of Uncertain Systems,” *Automatica*, vol. 159, paper 111332, 2024.
15. **F. Fotiadis**, K. G. Vamvoudakis, “Recursive Reasoning with Reduced Complexity and Intermittency for Non-Equilibrium Learning in Stochastic Games,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 11, pp. 8467-8481, 2023.
16. **F. Fotiadis**, K. G. Vamvoudakis, “Concurrent Receding Horizon Control and Estimation against Stealthy Attacks,” *IEEE Transactions on Automatic Control*, vol. 68, no. 6, pp. 3712-3719, 2023.
17. K. G. Vamvoudakis, **F. Fotiadis**, A. Kanellopoulos, N.-M. T. Kokolakis, “Nonequilibrium Dynamical Games: A Control Systems Perspective,” *Annual Reviews in Control*, vol. 53, pp. 6-18, 2022.
18. **F. Fotiadis**, G. A. Rovithakis, “Prescribed Performance Control for Discontinuous Output Reference Tracking,” *IEEE Transactions on Automatic Control*, vol. 66, no. 9, pp. 4409-4416, 2021.
19. **F. Fotiadis**, K. G. Vamvoudakis, “Detection of Actuator Faults for Continuous-Time Systems with Intermittent State Feedback,” *Systems & Control Letters*, vol. 152, pp. 104938, 2021.

Journal Preprints

20. S. Niu, A. Boulard, H. Wang, **F. Fotiadis**, A. Kurdila, A. L’Affitto, S. T. Paruchuri, K. G. Vamvoudakis, “Tailoring Reproducing Kernels for Optimal Control via Policy Iteration,” submitted.
21. **F. Fotiadis**, U. Topcu, “Adversarial Observability and Performance Tradeoffs in Optimal Control,” submitted.
22. N. Tukenmez, **F. Fotiadis**, J. M. Magalhães Júnior, K. G. Vamvoudakis, S. Bogosyan, “Intermittent Learning for Trajectory Tracking Control with an Experimental Validation on Microdrones,” submitted.

Conference Publications

23. A. Zongo, **F. Fotiadis**, U. Topcu, P. Wei, “Robust Multi-Agent Reinforcement Learning for Small UAS Separation Assurance under GPS Degradation and Spoofing,” to appear in *IEEE Conference on Decision and Control*, Honolulu, HI, 2026.
24. S. Niu, A. Boulard, H. Wang, **F. Fotiadis**, A. J. Kurdila, A. L’Affitto, M. Liu, S. T. Paruchuri, K. G. Vamvoudakis, “Greedy Methods via Reproducing Kernels for Approximations of Policy Iteration,” to appear in *IEEE Conference on Decision and Control*, Honolulu, HI, 2026.
25. J. Webb, B. Bjorkman, A. Diaz Gonzalez, A. Coursey, N. Dahle, K. Lemieux Mack, **F. Fotiadis**, G. Biswas, B. C. Ward, A. Taye, “Remote ID Spoofing-Aware Trajectory Planning for Small Unmanned Aerial Systems,” to appear in *AIAA/IEEE Digital Avionics Systems Conference*, Orlando, FL, 2026.
26. **F. Fotiadis**, B. M. Sadler, U. Topcu, “Coordinated UAV Beamforming and Control for Directional Jamming and Nulling,” to appear in *American Control Conference*, New Orleans, LA, 2026.
27. **F. Fotiadis**, Q. Rommel, G. Falco, U. Topcu, “Adversarial Pursuits in Cislunar Space,” *Proc. AIAA SciTech Forum*, Orlando, FL, 2026.
28. **F. Fotiadis**, Q. Rommel, B. M. Sadler, U. Topcu, “Optimal Satellite Maneuvers for Spaceborne Jamming Attacks,” *Proc. AIAA SciTech Forum*, Orlando, FL, 2026.
29. N. Tukenmez, J. Magalhães, **F. Fotiadis**, K. G. Vamvoudakis, A. Homaifar, “A Comprehensive Review of Trajectory Planning in Swarm Drones: Methods, Challenges, and Future Directions,” *Proc. AIAA SciTech Forum*, Orlando, FL, 2026.
30. I. Sharifi, M. Ghazanfari, A. Taye, P. Wei, M. H. Ahmed, H. T. Kim, M. Ghasemi, V. Gupta, N. Dahle, R. Canady, A. D. Gonzalez, A. Coursey, B. Bjorkman, C. L. Mack, B. C. Ward, X. Koutsoukos, G. Biswas, S. Hasan, I. Amundson, **F. Fotiadis**, U. Topcu, J. Lu, Q. A. Chen, N. Aryal, A. Ibrahim, K. Ras, A. Shirkhodaie, “A Survey of Security Challenges and Solutions for UAS Traffic Management (UTM) and Small Unmanned Aerial Systems (sUAS),” *Proc. AIAA SciTech Forum*, Orlando, FL, 2026.
31. M. Ghazanfari, I. Sharifi, A. Taye, P. Wei, B. C. Ward, X. Koutsoukos, G. Biswas, N. Dahle, R. Canady, A. D. Gonzalez, A. Coursey, B. Bjorkman, C. L. Mack, M. Ghasemi, M. H. Ahmed, H. T. Kim, V. Gupta, **F. Fotiadis**, U. Topcu, A. Chen, J. Lu, A. Shirkhodaie, N. Aryal, A.

- Ibrahim, K. Ras, S. Hasan, I. Amundson, “A Survey of Security Challenges and Solutions for Advanced Air Mobility and eVTOL Aircraft,” *Proc. AIAA SciTech Forum*, Orlando, FL, 2026.
32. K. G. Vamvoudakis, **F. Fotiadis**, T. Başar, V. Gupta, J. Poveda, M. Tang, M. Krstic, Q. Zhu, “Deception in Game Theory and Control: A Tutorial,” *Proc. American Control Conference*, pp. 2146-2163, Denver, CO, 2025.
 33. X. Liu, J. Li, **F. Fotiadis**, M. O. Karabag, J. Milzman, D. Fridovich-Keil, U. Topcu, “Policies with Sparse Inter-Agent Dependencies in Dynamic Games: A Dynamic Programming Approach,” *Proc. 24th International Conference on Autonomous Agents and Multiagent Systems*, pp. 2633-2635, Detroit, MI, 2025.
 34. **F. Fotiadis**, K. G. Vamvoudakis, “Input-Output Data-Driven Sensor Selection,” *Proc. IEEE Conference on Decision and Control*, pp. 4506-4511, Milan, Italy, 2024.
 35. **F. Fotiadis**, G. A. Rovithakis, K. G. Vamvoudakis, “Embedding Learning-based Optimal Controllers with Assured Safety,” *Proc. IEEE Conference on Decision and Control*, pp. 5266-5273, Milan, Italy, 2024.
 36. A. Kanellopoulos, **F. Fotiadis**, K. G. Vamvoudakis, H. Sandberg, “Unpredictable Switching for Cyber-Physical Security Against Worst-case Attackers,” *Proc. IEEE Conference on Decision and Control*, pp. 2361-2366, Milan, Italy, 2024.
 37. N. Tukenmez, **F. Fotiadis**, J. M. Magalhães Júnior, K. G. Vamvoudakis, S. Bogosyan, “Reward Drops in Learning-based Control with an Experimental Validation on Microdrones,” *Proc. IEEE Conference on Decision and Control*, pp. 3819-3824, Milan, Italy, 2024.
 38. **F. Fotiadis**, A. Kanellopoulos, K. G. Vamvoudakis, J. Hugues, “Poisoning Actuation Attacks Against the Learning of an Optimal Controller,” *Proc. American Control Conference*, pp. 4823-4828, Toronto, Canada, 2024.
 39. S. Athalye, **F. Fotiadis**, K. G. Vamvoudakis, J. Hugues, “An Output Feedback Game-Theoretic Approach for Defense against Stealthy GNSS Spoofing Attacks,” *Proc. American Control Conference*, pp. 3689-3694, Toronto, Canada, 2024.
 40. J. M. Magalhães Júnior, L. Zhai, **F. Fotiadis**, A. Kanellopoulos, K. G. Vamvoudakis, J. Hugues, “Real-Time and Experimental Reactive and Proactive Defense in a Multi-Agent Scenario,” *Proc. AIAA SciTech Forum*, Orlando, Florida, 2024.
 41. **F. Fotiadis**, K. G. Vamvoudakis, “A Physics-Informed Neural Networks Framework to Solve the Infinite-Horizon Optimal Control Problem,” *Proc. IEEE Conference on Decision and Control*, pp. 6008-6013, Marina Bay Sands, Singapore, 2023.
 42. **F. Fotiadis**, K. G. Vamvoudakis, “Game-Theoretic Deception Methods for Perfectly and Bounded Rational Stealthy Attackers,” *Proc. IEEE Conference on Decision and Control*, pp. 7956-7963, Marina Bay Sands, Singapore, 2023.
 43. **F. Fotiadis**, G. A. Rovithakis, “Input-Constrained Prescribed Performance Control for SISO Nonlinear Systems via Reference Relaxation,” *Proc. Mediterranean Conference on Control and Automation*, pp. 777-782, Limassol, Cyprus, 2023.

44. K. G. Vamvoudakis, **F. Fotiadis**, J. P. Hespanha, R. Chinchilla, G. Yang, M. Liu, J. S. Shamma, L. Pavel, "Game Theory for Autonomy: From Min-Max Optimization to Equilibrium and Bounded Rationality Learning," *Proc. American Control Conference*, pp. 4363-4380, San Diego, CA, 2023.
45. L. Zhai, **F. Fotiadis**, K. G. Vamvoudakis, J. Hugues, "Timing-Aware Resilience of Data-driven Off-policy Reinforcement Learning for Discrete-Time Systems," *Proc. American Control Conference*, pp. 2782-2787, San Diego, CA, 2023.
46. L. Zhai, A. Kanellopoulos, **F. Fotiadis**, K. G. Vamvoudakis, J. Hugues, "A Modular Approach to Verification of Learning Components in Cyber-Physical Systems," *Proc. AIAA SciTech Forum*, National Harbor, MD, 2023.
47. **F. Fotiadis**, K. G. Vamvoudakis, Z.-P. Jiang, "Data-Based Actuator Selection for Optimal Control Allocation," *Proc. IEEE Conference on Decision and Control*, pp. 4674-4679, Cancun, Mexico, 2022.
48. **F. Fotiadis**, A. Kanellopoulos, K. G. Vamvoudakis, J. Hugues, "Impact of Sensor and Actuator Clock Offsets on Reinforcement Learning," *Proc. American Control Conference*, pp. 2669-2674, Atlanta, GA, 2022.
49. L. Zhai, A. Kanellopoulos, **F. Fotiadis**, K. G. Vamvoudakis, J. Hugues, "Towards Intelligent Security for Unmanned Aerial Vehicles: A Taxonomy of Attacks, Faults, and Detection Mechanisms," *Proc. AIAA SciTech Forum*, 2022.
50. **F. Fotiadis**, K. G. Vamvoudakis, "Learning-based Actuator Placement for Uncertain Systems," *Proc. IEEE Conference on Decision and Control*, pp. 90-95, Austin, TX, 2021.
51. **F. Fotiadis**, C. Verginis, K. G. Vamvoudakis, U. Topcu, "Assured Learning-based Optimal Control subject to Timed Temporal Logic Constraints," *Proc. IEEE Conference on Decision and Control*, pp. 750-756, Austin, TX, 2021.
52. A. Kanellopoulos, **F. Fotiadis**, C. Sun, Z. Xu, K. G. Vamvoudakis, U. Topcu, W. E. Dixon, "Temporal Logic-based Intermittent, Optimal, and Safe Continuous-Time Learning for Trajectory Tracking," *Proc. IEEE Conference on Decision and Control*, pp. 1263-1268, Austin, TX, 2021.
53. **F. Fotiadis**, K. G. Vamvoudakis, "Recursive Reasoning for Bounded Rationality in Multi-Agent Non-Equilibrium Play Learning Systems," *Proc. IEEE Conference on Control Technology and Applications*, pp. 741-746, San Diego, CA, 2021.
54. A. Kanellopoulos, **F. Fotiadis**, K. G. Vamvoudakis, V. Gupta, "A Meta-Learning and Bounded Rationality Framework for Repeated Games in Adversarial Environments," *Proc. IEEE Conference on Decision and Control*, pp. 1640-1645, Jeju Island, Republic of Korea, 2020.
55. **F. Fotiadis**, A. Kanellopoulos, K. G. Vamvoudakis, "Constrained Differential Games for Secure Decision-Making against Stealthy Attacks," *Proc. American Control Conference*, pp. 4658-4663, Denver, CO, 2020.
56. F. Dimeas, **F. Fotiadis**, D. Papageorgiou, A. Sidiropoulos, Z. Doulgeri, "Towards Progressive Automation of Repetitive Tasks Through Physical Human-Robot Interaction," *Human Friendly Robotics: 10th International Workshop*, pp. 151-163, 2018.

Conference Preprints

57. C. Probine, Y. E. Bayiz, **F. Fotiadis**, S. Li, Y. Yang, U. Topcu, “Characterizing Opinion Evolution of Networked LLMs,” submitted.

Service Activities

Workshop Organizer: “Security and Safety of Space Systems: A Control-Theoretic Perspective,” IEEE Conference on Decision and Control, Honolulu, HI, 2026.

Tutorial Co-Organizer: “Deception in Game Theory and Control,” American Control Conference, 2025.

Program Committee Member: International Conference on Cyber-Physical Systems (Posters & Demos), 2025.

Session Co-Chair: Mediterranean Conference on Control and Automation (Nonlinear Control II), 2023.

Reviewer for Journals: IEEE Transactions on Automatic Control (>80), Automatica (>60), IEEE Transactions on Systems, Man, and Cybernetics: Systems (>20), IEEE Control Systems Letters (>20), IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Cybernetics, IEEE Transactions on Control of Network Systems, IEEE Transactions on Industrial Electronics, IEEE Transactions on Artificial Intelligence, IEEE Transactions on Control Systems Technology, IEEE Transactions on Aerospace and Electronic Systems, International Journal of Robust and Nonlinear Control, Systems & Control Letters, European Journal of Control, AIAA Journal of Guidance, Control, and Dynamics, Neurocomputing, Journal of Optimization Theory and Applications, Expert Systems with Applications, Optimal Control Applications and Methods, Applied Mathematics and Computation, The Aeronautical Journal, and others.

Reviewer for Conferences: IEEE Conference on Decision and Control (CDC) (>20), American Control Conference (ACC) (>10), European Control Conference (ECC), IFAC World Congress, Mediterranean Conference on Control and Automation (MED), IEEE Conference on Control Technology and Applications (CCTA), IFAC Conference on Modelling, Identification and Control of Nonlinear Systems, and others.